Workshop Syllabus: Blind 75 Problem Set

Duration: 11 weeks (1 session per week, 1.5 hours per session)

Target Audience: Graduates and job seekers with basic knowledge of DS&A

Week 1: Introduction & Arrays

Session 1: Introduction & Arrays

- **Topics:**
  - Why Blind 75?
  - Overview of the Blind 75 problem set
  - How to get the most out of this workshop series
  - Overview of what an average session will look like
  - Importance of problem-solving and interview preparation
  - Review of how to present technically in interviews
  - Technique/Tool: Hashmaps
  - Array problems: Two Sum, Valid Anagram, Contains Duplicate
- **Activities:**
  - Introduction and workshop goals
  - Guided practice solving 2-3 array problems
  - Collaborative practice solving additional array problem
  - Homework: Solve remaining array problems and review optimized solutions
  - Reflective journal: Key takeaways and challenges

Week 2: More Arrays

Session 2: Advanced Array Techniques

- **Topics:**
  - Technique/Tool: Bucket Sort and Prefix/Suffix Sum
  - Array problems: Product of Array Except Self, Top K Frequent Elements, Group Anagrams
- **Activities:**
  - Review of previous session's homework
  - Explanation of: Bucket Sort & Prefix/Suffix Sum Technique
  - Guided practice solving and discussing 2-3 advanced array problems
  - Collaborative practice solving an additional array problem
  - Homework: Solve remaining problems and review optimized solutions
  - Reflective journal: Progress and reflections

Week 3: Two Pointers

Session 3: Two Pointers

- **Topics:**
  - Introduction to two-pointer technique
  - Problems: Valid Palindrome, Two Sum II, Container With Most Water
- **Activities:**
  - Review of previous session's homework
  - Explanation of two-pointer technique
  - Guided practice solving and discussing 2-3 two-pointer problems
  - Collaborative practice solving additional two-pointer problem
  - Homework: Solve remaining two-pointer problems
  - Reflective journal: Thought process and improvements

Week 4: Sliding Window

Session 4: Sliding Window

- **Topics:**
  - Introduction to sliding window technique
  - Differences between sliding window vs two-pointer
  - Problems: Best Time to Buy and Sell Stock, Longest Substring Without Repeating Characters, Minimum Window Substring
- **Activities:**
  - Review of previous session's homework
  - Explanation of sliding window technique and how it contrasts with 2 pointer
  - Guided practice solving and discussing 2-3 sliding window problems
  - Collaborative practice solving additional sliding window problem
  - Homework: Solve remaining sliding window problems
  - Reflective journal: Techniques and insights

Week 5: Stacks and Queues

Session 5: Stacks and Queues

- **Topics:**
  - Introduction to stack and queue problems
  - Problems: Valid Parentheses, Daily Temperatures, Asteroid Collision
- **Activities:**
  - Explanation of stack and queue operations
  - Guided practice solving and discussing 2-3 stack/queue problems
  - Collaborative practice solving additional stack/queue problem
  - Homework: Solve remaining stack/queue problems

- Reflective journal: Challenges and strategies

Week 6: Linked Lists

Session 6: Linked Lists

- **Topics:**
  - Introduction to linked list problems
  - Problems: Reverse Linked List, Merge Two Sorted Lists, Reorder List
- **Activities:**
  - Explanation of linked list concepts
  - Guided practice solving and discussing 2-3 linked list problems
  - Collaborative practice solving additional linked list problem
  - Homework: Solve remaining linked list problems
  - Reflective journal: Learnings and difficulties

Week 7: Trees

Session 7: Trees

- **Topics:**
  - Introduction to tree problems
  - Review of BFS/DFS
  - Problems: Maximum Depth of Binary Tree, Same Tree, Invert/Flip Binary Tree
- **Activities:**
  - Explanation of tree concepts and traversal techniques
  - Guided practice solving and discussing 2-3 tree problems
  - Collaborative practice solving additional tree problem
  - Homework: Solve remaining tree problems
  - Reflective journal: Understanding tree structures

Week 8: Advanced Trees

Session 8: Advanced Trees

- **Topics:**
  - Problems: Binary Tree Maximum Path Sum, Validate Binary Search Tree, Kth Smallest Element In a BST
- **Activities:**
  - Explanation of advanced tree concepts
  - Guided practice solving and discussing 2-3 advanced tree problems
  - Collaborative practice solving additional advanced tree problem
  - Homework: Solve remaining advanced tree problems

    - Reflective journal: Reflect on advanced tree techniques

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Week 9: Graphs

Session 9: Graphs

- **Topics:**
    - Problems: Number of Islands, Clone Graph, Pacific Atlantic Water Flow
- **Activities:**
    - Explanation of graph concepts
    - Guided practice solving and discussing 2-3 graph problems
    - Collaborative practice solving additional graph problem
    - Homework: Solve remaining graph problems
    - Reflective journal: Review graph techniques

Week 10: Dynamic Programming

Session 10: Dynamic Programming

- **Topics:**
    - Introduction to dynamic programming (DP)
    - Problems: Climbing Stairs, Coin Change, Robber Problem
- **Activities:**
    - Explanation of dynamic programming concepts
    - Guided practice solving and discussing 2-3 DP problems
    - Collaborative practice solving additional DP problem
    - Homework: Solve remaining DP problems
    - Reflective journal: DP patterns and practice

Week 11: Review and Mock Interviews

Session 11: Review & Mock Interviews

- **Topics:**
    - Comprehensive review of key concepts and problem-solving techniques
    - Conducting mock technical interviews with participants
- **Activities:**
    - Review and discuss key problems from each category
    - Conduct mock interviews with feedback
    - Q&A session on interview strategies and experiences
    - Reflective journal: Final thoughts and future plans

Additional Materials and Resources

- **Pre-workshop Preparation:**
  - Share the Blind 75 problem set and prerequisite resources
- **Tools and Platforms:**
  - Neetcode
  - LeetCode for coding practice
  - Email or Discord for communication and support
- **Follow-up:**
  - Encourage continuous practice and progress sharing
  - Organize monthly coding challenges or follow-up sessions